

An Internship Report

Of

AWS Cloud

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)



By

Roll Number of Student:- 2410030701

Name of Student: SAMEER CHOUDHARY

Batch: 2024-28

CANDIDATE'S DECLARATION

I, SAMEER CHOUDHARY, do hereby declare that the internship report titled “AWS CLOUD” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: **SAMEER CHOUDHARY**

Roll No.: **2410030701**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **Cloud**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: **SAMEER CHOUDHARY**

Roll No.: **2410030701**

TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate	-
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	2
4.3 Problem Statement	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used	4
4.7 System Architecture	5
4.8 Methodology	6
4.9 Expected Outcomes	7
4.10 Certificates of Completion and Communication Proof	8
5. Bibliography/References	9

3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



भारतीय
शिक्षा मंत्रालय
NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

sameer choudhary

IILM University, Greater Noida

has successfully completed the 8-weeks

Cloud Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Curriculum Provided by:

aws academy



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4d16aeab40f45d7ca268

Student ID: STU69cc03d38d5901774978003



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter



Ref No: C16Y2026M071411006

Internship Offer Letter

Student Name : sameer choudhary
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : AWS Cloud
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU69cc03d38d5901774978003

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Cloud Foundations	Cloud concepts, economics, and billing while exploring AWS global infrastructure for foundational cloud knowledge and practical understanding.
Week 2	Cloud Foundations	Cloud security, networking, and content delivery, gaining practical knowledge to build secure and efficient cloud solutions.
Week 3	Cloud Foundations	AWS compute, storage, and database services, along with cloud architecture fundamentals for building scalable solutions.
Week 4	Cloud Foundations	Automatic scaling and monitoring in the cloud to optimize performance and manage resources efficiently.
Week 5	Cloud Architecting	Learn cloud architecture, IAM security, S3 storage, EC2 compute, RDS databases, VPC networking, and VPC peering.
Week 6	Cloud Architecting	Secure access using IAM and KMS, implement monitoring, elasticity, and high availability with EC2 and Route 53.
Week 7	Cloud Architecting	Build decoupled architectures using Amazon SQS and design serverless microservices with AWS Lambda.
Week 8	Cloud Architecting	Data engineering patterns, disaster recovery planning, and AWS Solutions Architect certification preparation.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



Scan to Verify



Curriculum provided by



Under the Patronage of



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “**AWS Cloud**”. The internship was carried out from April 2026 to Junet 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical understanding of AWS Cloud and their real-world, Cloud concepts. Over the course of the programme, the learner was exposed to industry-relevant tools and practices used to take Cloud concepts from a research or prototype stage to a fully deployed, monitored, and scalable production system.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, project documentation, and a final capstone project. On successful completion of all the requirements, including the final assessment test, a Virtual Internship Certificate was awarded by EduSkills Academy.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. The organization collaborates with leading technology companies and academic institutions to design industry-aligned virtual internship curricula that help engineering students strengthen their employability and gain practical exposure to emerging technology domains such as Artificial Intelligence, Cloud Computing, Cybersecurity, and MLOps, among others.

4.3 Problem Statement

A large number of engineering students study the theoretical foundations of computer networks, databases, and system design during their academic coursework, but very few get structured, hands-on exposure to deploying and managing infrastructure on a real cloud platform. Understanding cloud concepts in isolation is not sufficient; students also need practical exposure to designing secure, scalable, and highly available architectures using industry-standard tools. The problem addressed by this internship was to understand and practically apply core AWS Cloud concepts – ranging from global infrastructure, security, and networking to compute, storage, database services, and serverless architecture – in a structured, assessment-driven learning environment

4.4 Project Objectives

- To understand cloud computing concepts, economics, and billing, along with AWS global infrastructure.
- To gain practical knowledge of cloud security, networking, and content delivery on AWS.
- To learn AWS compute, storage, and database services, along with cloud architecture fundamentals.
- To understand automatic scaling and monitoring for optimising performance and managing resources.
- To learn cloud architecture design using IAM security, Amazon S3, Amazon EC2, Amazon RDS, and Amazon VPC (including VPC peering).
- To implement secure access using IAM and KMS, and to configure monitoring, elasticity, and high availability using EC2 and Route 53.
- To build decoupled architectures using Amazon SQS and design serverless microservices using AWS Lambda.
- To understand data engineering patterns and disaster recovery planning, and to prepare for the AWS Certified Solutions Architect – Associate certification.

4.5 Scope of the Project

The scope of this internship spans foundational and architectural aspects of AWS Cloud – starting with cloud concepts, economics, and global infrastructure, moving through security, networking, compute, storage, and database services, and culminating in cloud architecture design, secure access management, elasticity, decoupled and serverless architectures, and disaster recovery planning. The internship was conducted in a virtual, lab-based environment provided by EduSkills Academy in partnership with AWS Academy, and did not involve deployment on any live organisational production system.

4.6 Technologies and Tools Used

Table 4.1: AWS Services and Tools Used

Category	AWS Services / Tools
Identity & Access	AWS IAM, AWS KMS
Networking	Amazon VPC, VPC Peering, Amazon Route 53
Compute	Amazon EC2, Auto Scaling
Storage	Amazon S3
Database	Amazon RDS
Serverless & Messaging	AWS Lambda, Amazon SQS
Monitoring & Resilience	Amazon CloudWatch, Disaster Recovery Planning
Certification Track	AWS Certified Solutions Architect – Associate (preparation)

4.7 System Architecture

The general architecture followed during the AWS Cloud Internship can be summarized in the following layers:

- **Application Layer:** Provides the user interface through which users interact with the application.
- **Compute Layer:** AWS services such as **Amazon EC2 and AWS Lambda** are used to run application workloads.
- **Storage Layer:** **Amazon S3** is used to store files, datasets, and application resources.
- **Database Layer:** **Amazon RDS** is used to store and manage structured application data.

- **Security Layer: AWS IAM** manages users, roles, permissions, and secure access to AWS resources.
- **Networking Layer: Amazon VPC** provides a secure and isolated network environment for AWS resources.
- **Monitoring Layer: Amazon CloudWatch** is used for monitoring, logging, and tracking application performance.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, weekly assessments, and project submissions, culminating in a Final Assessment Test, as summarised in Table 4.1.

Table 4.1: Week-wise Internship Methodology

Week	Module	Description
1	Cloud Foundations	Cloud concepts, economics, and billing, along with exploring AWS global infrastructure for foundational cloud knowledge and practical understanding.
2	Cloud Foundations	Cloud security, networking, and content delivery, gaining practical knowledge to build secure and efficient cloud solutions.
3	Cloud Foundations	AWS compute, storage, and database services, along with cloud architecture fundamentals for building scalable solutions.
4	Cloud Foundations	Automatic scaling and monitoring in the cloud to optimize performance and manage resources efficiently.
5	Cloud Architecting	Cloud architecture, IAM security, S3 storage, EC2 compute, RDS databases, VPC networking, and VPC peering.
6	Cloud Architecting	Secure access using IAM and KMS, and implementation of monitoring, elasticity, and high availability with EC2 and Route 53.
7	Cloud Architecting	Building decoupled architectures using Amazon SQS and designing serverless microservices with AWS Lambda.
8	Cloud Architecting	Data engineering patterns, disaster recovery planning, and AWS Solutions Architect certification preparation.

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and assignments. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across weekly assessments, project submissions, and the final test. No stipend was associated with this internship.

4.9 Expected Outcomes

- Ability to explain core AWS Cloud concepts, global infrastructure, and cloud economics/billing.
- Practical understanding of cloud security, networking, and content delivery on AWS.
- Hands-on familiarity with AWS compute (EC2), storage (S3), and database (RDS) services.
- Ability to design secure architectures using IAM, KMS, VPC, and VPC peering.
- Understanding of elasticity, auto scaling, and high availability using EC2 and Route 53.
- Capability to design decoupled and serverless architectures using Amazon SQS and AWS Lambda.
- Awareness of data engineering patterns and disaster recovery planning on AWS.
- A structured foundation for pursuing the AWS Certified Solutions Architect – Associate certification.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 10-weeks AICTE – EduSkills Virtual Internship Program in AWS Cloud, are attached in Chapter 3 of this report.

download

External

Inbox



EduSkills Foundation 11 Jun

to me ▾



EduSkills Foundation

Internship Completion Certificate

Dear **sameer choudhary**,

 **Congratulations!**

You have successfully completed **10 weeks AWS Cloud** internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.



Steps to Download Your Certificate:

1. Login at: <https://aikte-internship.eduskillsfoundation.org/>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, “AICTE – EduSkills Virtual Internship Program – Cloud Virtual Internship (AWS Cloud),” Internship Offer Letter, Ref No. C16Y2026M071411006, 2026.
2. EduSkills Academy, “Certificate of Virtual Internship – Cloud Virtual Internship,” Certificate ID: 4d16aeab40f45d7ca268, Student ID: STU69cc03d38d5901774978003, 2026.
3. AWS Academy Curriculum, Amazon Web Services, <https://aws.amazon.com/training/awsacademy/>
4. Amazon Web Services Documentation, <https://docs.aws.amazon.com/>
5. AWS Identity and Access Management (IAM) Documentation, <https://docs.aws.amazon.com/iam/>
6. Amazon Virtual Private Cloud (VPC) Documentation, <https://docs.aws.amazon.com/vpc/>
7. AICTE National Internship Portal, <https://internship.aicte-india.org/>